Name:

Class: Date:

GCSE TOPIC TEST

PHYSICS

H

Higher Tier

4.8 Space Physics

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
TOTAL	

Materials

For this paper you must have:

- a ruler
- a scientific calculator
- the Physics Equation sheet

Instructions

- Use black ink or black ball-point pen.
- Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- In all calculations, show clearly how you work out your answer.

Information

- The maximum mark for this paper is 37.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.
- You are reminded of the need for good English and clear presentation in your answers

01.1 How are planets and moons similar?

Tick **two** boxes.

- ☐ Their mass is about the same
- ☐ Their orbits are circular
- ☐ Their surfaces are the same colour
- ☐ They are similar in diameter
- ☐ They do not emit visible light

[2 marks]

01.2 **Table 1** shows data about five planets.

Table 1

Planet	Mean distance from the Sun in millions of kilometres	Mean surface temperature in °C
Earth	150	+22
Mars	228	−48
Jupiter	778	**X**
Saturn	1430	−178
Uranus	2870	−200

How does the mean surface temperature of the planets in **Table 1** change as the mean distance from the Sun increases?

____

____

[1 mark]

01.3 Predict the mean surface temperature of Jupiter (**X**) in **Table 1** above.

Mean surface temperature of Jupiter = ____________ °C

[1 mark]

01.4 Five of the planets in the solar system are given in **Table 1** above.

How many other planets are there in the solar system?

Tick **one** box.

- ☐ Two
- ☐ Three
- ☐ Four
- ☐ Five

[1 mark]

02 **Table 2** gives information about some of the planets in our solar system. The planets are in order of increasing distance from the Sun.

Table 2

Planet	Time to orbit the Sun in years
Mercury	0.2
Venus	0.6
Earth	1.0
Mars	
Jupiter	12.0

Estimate how many years it takes Mars to orbit the Sun.

____________ years

[1 mark]

03.1 The Big Bang theory is one way to explain the origin of the universe.

How does the Big Bang theory describe the universe when it began?

Tick **one** box.

- ☐ Very big and very dense
- ☐ Very big and extremely hot
- ☐ Very dense and extremely hot
- ☐ Very small and extremely cold

[1 mark]

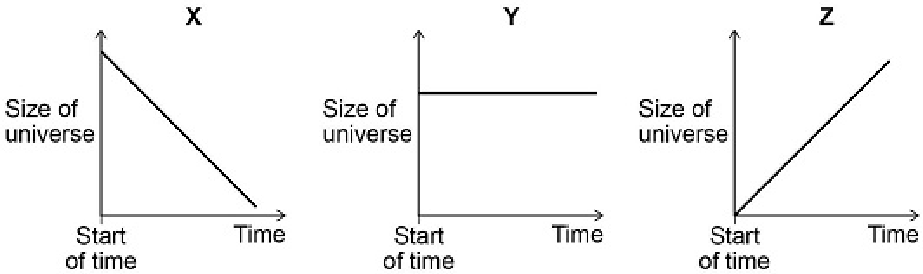
03.2 Which statement about the Big Bang theory is correct?

Tick **one** box.

- ☐ Scientists have proved that the theory is correct
- ☐ Scientific evidence supports the theory
- ☐ There is no other way to explain the origin of the universe

[1 mark]

03.3 The graphs below show three ways that the size of the universe may have changed with time.



Which graph would the Big Bang theory suggest is correct?

Tick **one** box.

☐ X

☐ Y

☐ Z

Give a reason for your answer.

[2 marks]

03.4 Most galaxies are moving away from the Earth. Scientists can determine the speed of a galaxy by observing the light from the galaxy.

Tick **one** box to complete the sentence.

When scientists observe the light from distant galaxies, they observe an increase in the

- ☐ frequency
- ☐ speed
- ☐ wavelength

of light from those galaxies.

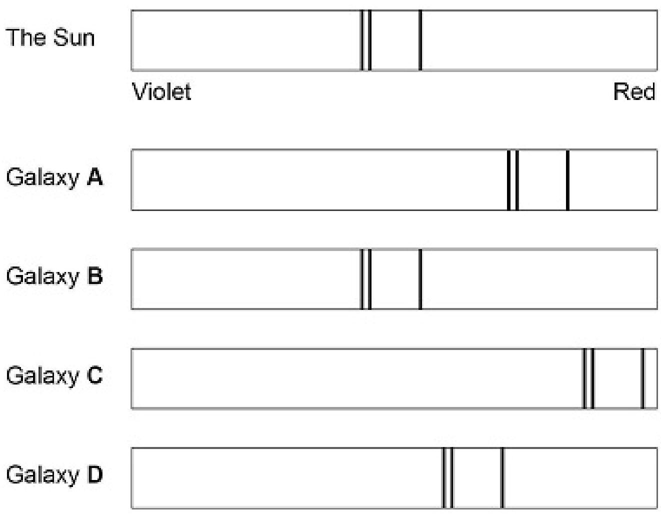
[1 mark]

03.5 The light spectra from stars and galaxies include dark lines.

The lines have the same pattern.

Figure 1 shows the light spectrum from the Sun and from four galaxies.

Figure 1



Which galaxy is moving the fastest away from the Earth?

Tick **one** box.

☐ A

☐ B

☐ C

☐ D

[1 mark]

03.6 Which galaxy is the furthest away from the Earth?

Tick **one** box.

☐ A

☐ B

☐ C

☐ D

[1 mark]

04.1 Complete the sentences.

The Sun is a stable star. This is because the forces pulling inwards caused by ____________ are in equilibrium with the forces pushing outwards caused by the energy released by nuclear ____________.

[2 marks]

04.2 Some stars are much more massive than the Sun.

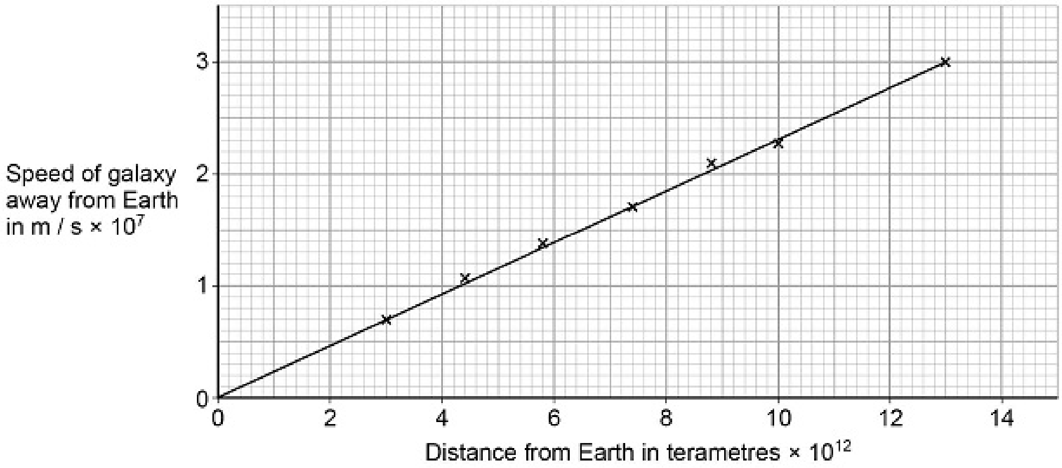
Describe the life cycle of stars much more massive than the Sun, including the formation of new elements.

[6 marks]

05.1

Figure 2 shows data scientists have calculated from measurements of red-shift.

Figure 2



Describe the relationship between the speed of a galaxy and the distance the galaxy is from the Earth.

[1 mark]

05.2

The Big Bang theory suggested that gravity would slow the rate at which galaxies move away from the Earth.

New observations suggest that distant galaxies are moving away from the Earth at an increasingly fast rate.

What do the new observations suggest is happening to the universe?

[1 mark]

05.3 New observations and data that do not fit existing theories should undergo peer review.

Give **one** reason why peer review is an important process.

[1 mark]

06 **Table 2** gives information about some of the planets in our solar system. The planets are in order of increasing distance from the Sun.

Table 2

Planet	**Time to orbit the Sun in years**
Mercury	0.2
Venus	0.6
Earth	1.0
Mars	
Jupiter	12.0

Calculate how many times Venus will orbit the Sun in 9 years.

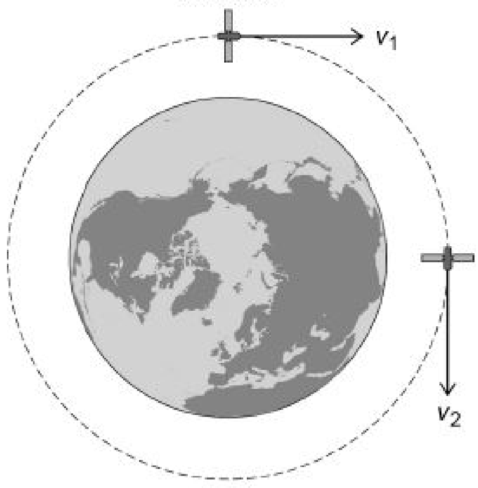
In 9 years Venus will orbit the Sun ____________ times.

[2 marks]

07.1 A satellite is in a circular orbit around the Earth.

Figure 3 shows the velocity of the satellite at two different positions in the orbit.

Figure 3

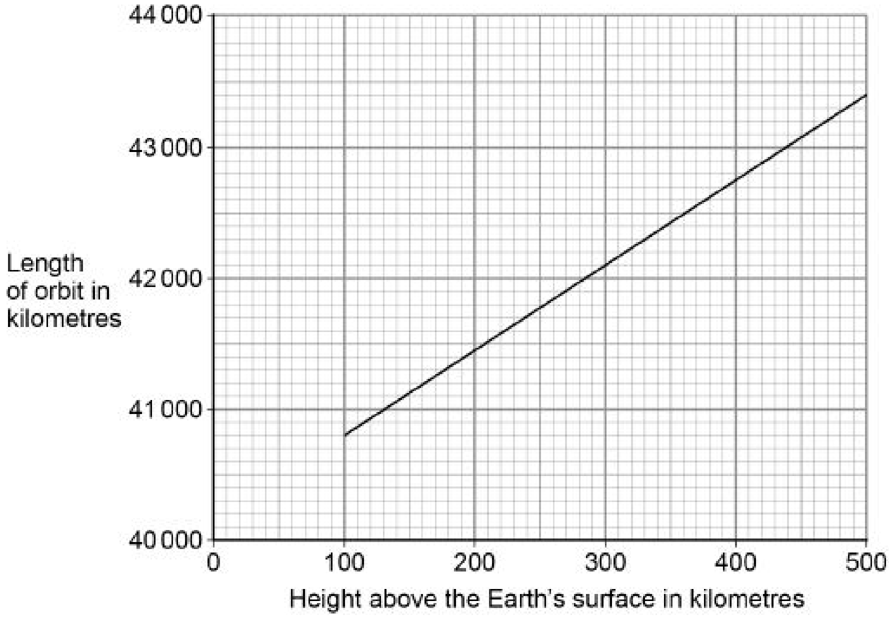


Explain why the velocity of the satellite changes as it orbits the Earth.

[3 marks]

07.2 **Figure 4** shows how the length of a satellite orbit depends on the height of the satellite above the Earth's surface.

Figure 4



A satellite orbits 300 km above the Earth's surface at a speed of 7.73 km/s.

Calculate how many complete orbits of the Earth the satellite will make in 24 hours.

Number of complete orbits =

[5 marks]

07.3 In 1772, an astronomer called J Bode developed an equation to predict the orbital radii of the planets around the Sun.

Table 3 shows Bode's predicted orbital radii and the actual orbital radii for the planets that were known in 1772.

Table 3

Planet	**Predicted orbital radius in millions of kilometres**	**Actual orbital radius in millions of kilometres**
Mercury	60	58
Venus	105	108
Earth	150	150
Mars	240	228
Jupiter	780	778
Saturn	1500	1430

The predicted data can be considered to be accurate.

Give the reason why.

[1 mark]

07.4 J Bode used his equation to predict the existence of a planet with an orbital radius of 2940 million kilometres.

The planet Uranus was discovered in 1781.

Uranus has an orbital radius of 2875 million kilometres.

Explain why the discovery of Uranus was important.

[2 marks]